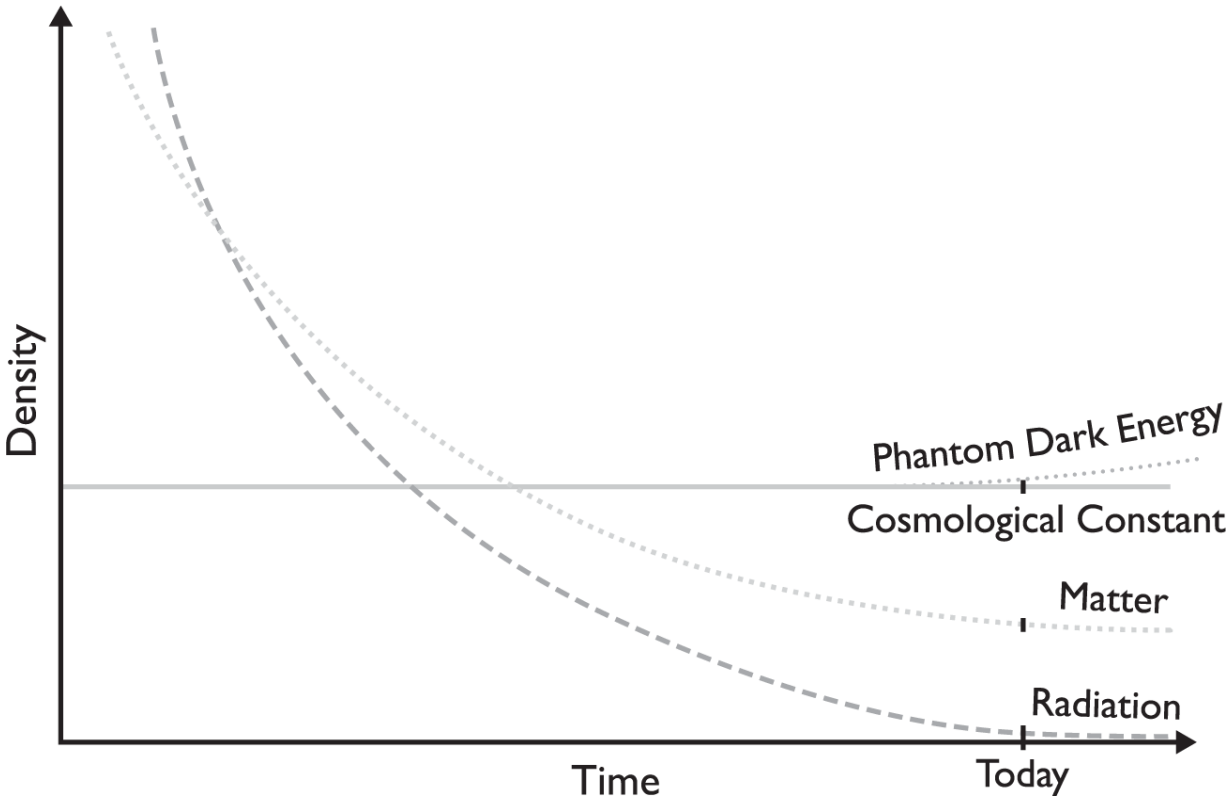


Figure 14: Evolution of dark energy in the form of a cosmological constant or in the form of phantom dark energy, compared with matter and radiation. While a cosmological constant keeps a constant density as the universe expands, in the case of phantom dark energy, the density increases.

Caldwell and his colleagues went ahead and calculated constraints based on a full range of possibilities for w. Not only did they find that values below -1 were perfectly consistent with the data, they also found, through a simple, straightforward calculation, that if w is even infinitesimally lower than -1, dark energy will tear the entire universe apart, and it will do so in a finite, calculable time.

I just want to pause for a moment to say that this paper, titled “Phantom Energy: Dark Energy with w < -1 Causes a Cosmic Doomsday,” is one of my absolute favorite papers in physics. It’s not often that you get to take some very mild-seeming alteration to the current perspective, shifting a parameter down by a minuscule amount, and find out that this DESTROYS THE ENTIRE UNIVERSE. Not only that, it gives you a way to calculate exactly *how* the universe will be destroyed, and when, and what it will look like when it all goes down.

Which is as follows.